INDIAN PATENT APPLICATION // DESIGN SPECIFICATION

Ultra-High Temperature Plasma Arc Reactor for Single-Stage Smelting & Precious Metal Recovery from Printed Circuit Board (PCB) E-Waste

Inventors: EWEM Technical Research Group · **Filing Ref:** 2026/EWEM/PAT-04

ABSTRACT

A thermodynamic reactor system and method for recovering precious and strategic metals (Au, Ag, Cu, Pd, Pt) from shredded electronic waste PCBs using a non-transferred thermal plasma torch. Operating at thermal core temperatures between 5,000 °C and 15,000 °C under an argon-nitrogen inert atmosphere, the system prevents dioxin and furan genesis. Volatilized organics are converted into high-calorific syngas, while base metals and molten precious metals separate by gravitational density from vitrified, non-leachable silicate slag, achieving >95% recovery yield.

TECHNICAL SPECIFICATIONS & PROCESS PARAMETERS

Parameter	Specification Value	Functional Mechanism
Plasma Core Temperature	5,000 °C – 15,000 °C	Non-transferred DC plasma torch with tungsten cathode
Molten Bath Operating Temp	1,450 °C – 1,600 °C	Molten copper collector phase for precious metal dissolution
Gas Retention Time	> 2.0 seconds	Thermal cracking preventing aromatic halogenated hydrocarbons
Precious Metal Yield	> 95.4% (Au, Ag, Pd, Pt)	Gravitational separation and multi-stage tapping
Slag Vitrification State	Zero-Leach Silicate	Compliant with US EPA TCLP toxicity limits